Consent and Info

Thank you for your interest in participating in our study!

Please carefully read the consent form in this survey. Please continue with the study once you have completed the consent form. You may complete this at any time and on ANY device; however, we recommend a computer for ease of reading.

If you do not want to participate, your responses will not be used in our study and will be destroyed.

If you have any questions about this study, please contact [redacted]

The study associated with this form has been exempted by the [redacted] IRB under [redacted]

Principal Investigator: [redacted]

Faculty Advisors:

- [redacted]
- [redacted]

You are invited to participate in a research study about how requirements engineering training affects AI researcher or practitioner ability to identify stakeholder requirements in an AI for Social Impact context. The results of this study will help the AI community improve training for individuals involved in AI for Social Impact development and ultimately improve the efficacy of AI4SI deployment.

If you agree to be part of the research study, you will be asked to complete the following:

- You will complete an online eligibility and consent form at your own time and on your own electronic device. (3-5 minutes)
- You will complete one of three types of training, selected at random: no training, requirements engineering training, or a modified requirements engineering training (0-10 minutes).
- You will then be exposed to **four** short written scenarios of stakeholders, answer comprehension questions about the stakeholders, and select components to build an AI system for each stakeholder scenario (13-14 minutes *each*)
- You will answer a set of questions about your demographics (4-5 minutes)

There is no strict time limit for completing the survey. Most participants in our pilot study took approximately 60 to 75 minutes. We suggest spending about 13-14 minutes on each stakeholder scenario, which would result in 60-75 minutes. You are welcome to take more or less time based on your pace, but the compensation amount will remain the same.

You can do the survey in multiple sessions. Your progress should be saved if you use the same browser. Consider keeping the survey tab open if you plan to take a break and come back.

Participant responses will only count towards survey submission if they demonstrate a good faith effort and meet a minimum level of engagement and understanding, as determined by the research team.

Benefits of the research:

[redacted]

You may be assigned to a training condition that introduces concepts from Requirements Engineering. While some participants may find this information helpful, not all participants will receive the same training or experience the same level of benefit. However, others may benefit from the knowledge gained from this study.

In modern AI development and AI education, it is of great interest to AI companies, institutions, and researchers to improve the training of AI developers. One part is to understand why AI deployments fail in the real world, especially when the AI systems are working with real people in real situations. The purpose of this study is to learn more about the project development processes involved in this task. You are being asked to participate in this study because it requires participants with some computer science knowledge, such as data structures, as well as basic knowledge of AI systems, to perform a number of tasks in a survey. If you do not have the required fundamental knowledge, you cannot participate in this study.

Risks and discomforts:

There may be some risk of temporary boredom while taking the survey. This risk is minimal and you are able to stop participating in the study at any time for any reason without consequence.

In addition, data breaches may result in the disclosure of personal data and vulnerability. We will store personal identifiers such as your name, email, and address. Additionally, research data collected from you will be de-identified after collection and stored on an encrypted volume. Your data will not be connected to any identifiers. Lastly, all paper surveys and consents will be stored in a locked office and locker by the Principal Investigator. Electronic data will be stored in a secure server managed by the faculty advisor.

Compensation:

You will receive a $30 virtual gift card at the end of the study if you demonstrate a good faith effort and meet a minimum level of engagement and understanding as determined by the research team.

Participants who do not submit the study, or do not demonstrate good faith effort will not be compensated.

If you receive any payments for taking part in this study, [redacted] finance department will need your name and address for tax reporting purposes. In a calendar year if: 1) your payments total greater than $400 for this study or 2) if you receive payments of greater than $400 for being in more than one study, [redacted] finance department will also require your Social Security Number for tax reporting purposes. If you do not wish to provide your Social Security Number, you may continue to participate in research studies, but you will not be able to receive payment for the remainder of the calendar year.

Participating in this study is completely voluntary. Even if you decide to participate now, you may change your mind and stop at any time. You may choose not to answer any survey question or continue with survey completion for any reason. However, you will not be compensated if you do not fully complete the survey.

Participants data and identifiable information such as name and address will be separated. In addition, participants' data will be stored in a secure server or in a secure locked location in which only the principal investigator and faculty advisor will have access. Identifiable information will be destroyed. Your research information will be stored electronically on the cloud; the term “cloud” refers to large computers located in different parts of the world where individuals may keep and remotely access their personal and professional files. Each cloud service has its own policies and methods for preventing unauthorized individuals from accessing files stored on their cloud servers. The cloud service used to store files associated with this study meets [redacted]

Information collected in this project may be shared with other researchers, but we will not share any information that could identify you.

[redacted]

As part of their review, the [redacted]

[redacted] has determined that this study is no more than minimal risk and exempt from on-going IRB oversight.

Consent to Participate in the Research Study

By signing this document, you are agreeing to be in this study. Make sure you understand what the study is about before you sign.

By signing, you verify that you meet the requirements of the study:

1. **You are older than 18 years of age**
2. **You have taken or are currently enrolled in a Data Structures & Algorithms course** [redacted]
3. Recommended that you are familiar with one or more of the fundamentals of AI (e.g., machine learning, reinforcement learning, agents, optimization etc.)

We will give you a copy of this document for your records and we will keep a copy with the study records. If you have any questions about the study after you sign this document, you can contact the study team using the information in Section 9 provided above.

I understand what the study is about and my questions so far have been answered. I agree to take part in this study.

(If you understand, write your name below)

Today's Date (mm/dd/yyyy):

(sign your name)

SIGN HERE

×

clear

SE Training

You will be receiving Requirements Engineering training. Requirements Engineering is the process of identifying, documenting, and managing what a system should do. It is a key part of both systems and software engineering. In this training, you'll learn how to communicate with stakeholders or customers to better understand their needs and define the right software solution for their specific context.

The first part of the survey includes a short training on requirements engineering concepts, including how to write user stories, define acceptance criteria, and use the MoSCoW method to categorize requirements. You'll go through a series of short text slides, with a few quick questions to check your understanding.

Please do not google, look up any terms or examples, or use generative AI (e.g. ChatGPT, etc.). Just try your best. The goal of the survey is NOT to get the answers all correct, but to be exposed to the material.

I understand and agree:

SIGN HERE

×

clear

We will start with user stories:

Who is a user?

In the context of user stories from requirements engineering, **a user is typically defined as an individual or role who interacts directly with the system to achieve a specific goal.** This includes:

- End users of the system
- Stakeholders who interact with or are affected by the system's functionality
- Other systems or agents that trigger system responses (e.g. API clients)

What is a User story?
A user story is a short, simple description of something a user wants to do, written from the user's perspective. It captures:

- WHO the user is
- WHAT they want to do
- WHY it matters to them

The general format is:

- **As a [type of user], I want [to do something], so that [I achieve this goal].**

This format helps center the user's role, action, and motivation, and keeps the conversation focused on problems, not just features or interface ideas.

User Stories are used in design and development to help teams:

- Focus on what matters most to the user
- Avoid premature design or technical decisions
- Clarify core problems and stakeholder goals
- Enable prioritization based on real needs

They’re especially helpful in complex systems or when working with many stakeholders. Writing user stories helps you better understand users’ mental models, challenges, and values.

Now we will show you some examples of what user stories can look like:

Example User Story: **As a [student], I want [to track my assignment deadlines], so that [I stay organized].**

This story captures a real user goal without diving into how the system will do it. It avoids specifying a calendar, alert system, or dashboard, focusing instead on **what the user wants and why.**

What makes a bad User Story?

- Too focused on the solution or not reflecting an achievable user goal:
  - As a [student], I want [the app to have a calendar feature], so that [it’s more high-tech]
- No clear action, task, or problem is specified
  - As a [student], I want [the app to be easy to use], so that [it’s not confusing].
- Written from the perspective of the developer rather than the user
  - As a [developer], I want [the app to support notifications], so that [we can increase engagement].

Choose the best user story for this scenario:

Hey, I'm Sarah. As a working parent, I rely on a mobile banking app to manage my finances on the go. I often check my account between meetings and sometimes need to quickly transfer money, pay bills, or view recent transactions. I value speed and simplicity because I don't have time to navigate through complex menus.

- ○ As a user, I want to see my transactions on the home screen, so that I can quickly check my spending.

- ○ As a user, I want to manage my finances better, so that I can feel more in control.
- ○ As a user, I want a search bar on every screen, so that I can find what I need.
- ○ As a user, I want notifications for my transactions, so that I know what's going on.

Now we'll move onto a more complicated example： Matt, the hospital administrator talks about his situation.

Hey, I'm Matt. I'm a hospital administrator overseeing the residency program at a hospital called Mercy Grace Hospital. We're one of the largest hospitals in the country and are currently ranked number one for cancer treatment and patient outcomes. A major contributor to our success is our focus on the early detection of tumors. We shuttle a lot of money into tech infrastructure that supports our imaging department and research labs, so we always have the latest computers and other hardware.

Our doctors in training are highly skilled at reviewing MRI images and identifying tumors at early stages. This expertise is largely thanks to a specialized training program we developed, where we use archived MRI scans from previous patients. Doctors in training review these images, attempt to diagnose any tumors, and then have their findings and notes verified by senior physicians.

Because of this, we've built a vast repository of MRI images paired with detailed doctor annotations. For example, we have extensive collections of spine MRI scans showing tumors, with doctor notes describing the tumor size, location (such as thoracic or lumbar spine), morphology, degree of invasion into surrounding tissues, and critical indicators like edema or signal intensity. Any type of cancer you can name, we have notes for it. These detailed annotations help our trainees understand not just whether a tumor is present, but also the nuanced clinical reasoning behind each diagnosis.

However, recently, a significant number of patients across our hospital have declined to consent to having their imaging data stored for educational use. They're worried about patient data safety, which makes sense. As a result, we are no longer allowed to pull historical MRI images and use them directly in training sessions for our doctors in

training.

We are now in urgent need of a new solution to continue training residents at the same high standard and to maintain our reputation as the nation's leading cancer center. It's important to note that we still have backend access to previous patient MRI images and corresponding doctor notes for internal reference and quality improvement purposes — but we can no longer use them directly for training-focused activities involving doctors in training.

I've been thinking about whether or not we could build a secure training platform that displays reconstructed or modified versions of existing scans, with identifying details removed and images altered enough to meet compliance rules, but still clinically useful. A well-designed software system could let us simulate diagnostic exercises using variations of real cases, drawing on the metadata, annotations, and visual patterns we've already collected. It could allow residents to walk through structured diagnosis pathways, see clinical justifications, and receive feedback automatically based on stored expert annotations.

We could also build in a case authoring tool for senior physicians to input de-identified, illustrative cases when they have time, helping us expand the training database in a controlled way. We had some doctors say they could volunteer to give feedback on some of these modules, though I doubt they'll have time to do so unless the platform is really easy to use.

Anyways, we're aware that another hospital across the country is beginning to explore a similar approach to their training. We definitely want to create and retain our own proprietary method, with the longer-term goal of licensing our training system and intellectual property to other institutions.

Your task is to generate user stories from the previous scenario. Consider these pre-writing prompts before you begin:

Pre-writing prompts: As you write user stories, think about:

- Who might be harmed or excluded if this system is poorly designed?

- What kind of resources (time, data, workforce, etc.) do the stakeholders have?
- What are really the main issues the stakeholders are facing?
- What is the stakeholder's ultimate goal?

Start by identifying distinct users and their core needs. For this example you might identify:

- Hospital administrator
- Doctors in training
- Patients
- IT or compliance officer

Then follow the basic user story template:

- As a [type of user], I want [to do something], so that [I achieve this goal].

For example, one user story you might come up with for the hospital administrator might be:

- As a [hospital administrator] , I want [to continue training our doctors with high quality training materials], so that [we can continue to ensure quality patient care].

Other example user stories for this scenario:

- As a [doctor in training], I want [to access MRI cases], so that [I can practice diagnosis].
- As an [IT or compliance officer], I want to [audit logs], so that [we can ensure HIPAA compliance and patient data safety].

Each user can have multiple user stories, especially if they interact with the system in different ways or have several needs. While thinking about the pre-writing prompts, you may come up with different categories within each user.

Come up with your own user stories for each stakeholder that you identify as relevant in the scenario. There is no limit to the number of user stories you can write. Place one user story per line, and follow the template of "As a type of user, I want to do something, so

that I achieve this goal."

There is no right or wrong. Try your best to **identify the key issues that the stakeholders are facing and what type of resources they have.**

▶ 📌 **Click here to expand and see the scenario from Matt again** 📌

At this point, we've captured what the user cares about. Now, we can think about how to create a solution that will fulfill the user's needs.

To do this, we can use acceptance criteria.

Once you've written a user story, the next step is to define acceptance criteria. Acceptance criteria are clear, testable conditions that must be met for the story to be considered complete. Think of acceptance criteria as a checklist for validating the story. They define what the system must do to fulfill the user's need.

Acceptance criteria can be written for each user story in the form of "Given," "When," Then."

Each criterion should:

- Be testable (manual or automated)
- Be clear and concise
- Provide user perspective

Each criterion should avoid:

- Criteria that rely on back-end specifics (e.g., "Uses Firebase to store hours")
- Vague expectations (e.g., “Should be user-friendly”)
- UI-specific mandates too early (e.g., “The hours must be displayed in a red pop-up modal”)

Example User Story:
As a [patient], I want [to control whether my medical images are used in training], so that [I can protect my privacy and feel safe about how my data is handled].

Example Acceptance Criteria:

- Given I am completing hospital consent forms, When I am asked about the use of my medical data for educational purposes, Then I must have a clear and separate option to either consent or decline.
- Given I have opted in, When my data is accessed by the training system, Then I must receive a notification in my patient portal that summarizes when and how it was used.

Write a few more acceptance criteria for the example user story:

▶ 📌 **Click here to expand and see the scenario from Matt again** 📌

Finally, we can use the MoSCoW method to categorize and prioritize our acceptance criteria.

The **MoSCoW** Method is a simple way to prioritize requirements by organizing them into four categories: Must have, Should have, Could have, and Won’t have. The name comes

from the capitalized letters—M, S, C, and W—with the "o"s added just to make it pronounceable.

This method helps teams focus on what really matters. **Each person may have a different categorization.** The MoSCoW method is intended to kickstart thinking about prioritization, and there is no one size fits all answer.

- Must have items are critical. Without them, the product or feature simply doesn't work, doesn't deliver value, or is unsafe or illegal.
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After developing acceptance criteria, we can categorize them using the MoSCoW method to determine which features to work on first.

Categorize the following example acceptance criteria from the same user story using the MoSCoW method as you see fit:

As a [patient], I want [to control whether my medical images are used in training], so that [I can protect my privacy and feel safe about how my data is handled].

	Must Have	Should Have	Could Have	Won't Have
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Given I am unsure about my current status, When I access the patient portal, Then I must see a clear and up-to-date record of my consent choice and	○	○	○	○

Must Have Should Have Could Have Won't Have

a brief explanation
of what that means.

AI Training

You will be receiving modified Requirements Engineering training. Requirements Engineering is the process of identifying, documenting, and managing what a system should do. It is a key part of both systems and software engineering. In this training, you'll learn how to communicate with stakeholders or customers to better understand their needs and define the right software solution for their specific context.

The first part of the survey includes a short training on requirements engineering concepts, including how to write user stories, define acceptance criteria, and use the MoSCoW method to categorize requirements. You'll go through a series of short text slides, with a few quick questions to check your understanding.

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Because of this, we’ve built a vast repository of MRI images paired with detailed doctor annotations. For example, we have extensive collections of spine MRI scans showing tumors, with doctor notes describing the tumor size, location (such as thoracic or lumbar spine), morphology, degree of invasion into surrounding tissues, and critical indicators like edema or signal intensity. Any type of cancer you can name, we have notes for it. These detailed annotations help our trainees understand not just whether a tumor is present, but also the nuanced clinical reasoning behind each diagnosis.

However, recently, a significant number of patients across our hospital have declined to consent to having their imaging data stored for educational use. They’re worried about patient data safety, which makes sense. As a result, we are no longer allowed to pull historical MRI images and use them directly in training sessions for our doctors in training.

We're now in urgent need of a new solution to continue training residents at the same

high standard and to maintain our reputation as the nation's leading cancer center. It's important to note that we still have backend access to previous patient MRI images and corresponding doctor notes for internal reference and quality improvement purposes, but we can no longer use them directly for training-focused activities involving doctors in training.

I've also been thinking about whether we could automate the early detection process itself. If there's a way to build a system that reliably flags potential tumors in real time, it could both support our doctors and help us catch more cases even earlier.

How can we continue to effectively train our doctors under these new restrictions? Is it possible to generate highly realistic, medically accurate synthetic MRI images that mimic real patient scans for training purposes? These images would need to accurately represent real tumor characteristics, including size, location, shape, tissue interactions, and imaging features to make sure that training remains clinically rigorous. I want to get a ton of these images so I can start rolling them out into training as soon as we get them. We had some doctors that said they could volunteer to give feedback on some of these systems, but I doubt that they'll have time to do so since they're already so busy.

Anyways, we're aware that another hospital across the country is beginning to explore a similar approach to their training. We definitely want to create and retain our own proprietary method, with the longer-term goal of licensing our training system and intellectual property to other institutions.

Your task is to generate user stories from the previous scenario.

Consider these pre-writing prompts before you begin:

Pre-writing prompts: As you write user stories, think about:

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- Hospital administrator
- Doctors in training
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Then follow the basic user story template:

- As a [type of user], I want [to do something], so that [I achieve this goal].

For example, one user story you might come up with for the hospital administrator might be:

- As a [hospital administrator] , I want [to continue training our doctors with high quality training materials], so that [we can continue to ensure quality patient care].

Other example user stories for this scenario:

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There is no right or wrong. Try your best to **identify the key issues that the stakeholders are facing and what type of resources they have.**

▶ 📌 **Click here to expand and see the scenario from Matt again** 📌

At this point, we've captured what the user cares about. Now, we can think about how to create a solution that will fulfill the user's needs.

To do this, we can use acceptance criteria.

Once you've written a user story, the next step is to define acceptance criteria. Acceptance criteria are clear, testable conditions that must be met for the story to be considered complete. Think of acceptance criteria as a checklist for validating the story. They define what the system must do to fulfill the user's need.

Acceptance criteria can be written for each user story in the form of "Given," "When," Then."

Each criterion should:

- Be testable (manual or automated)
- Be clear and concise
- Provide user perspective

Each criterion should avoid:

- Criteria that rely on back-end specifics: "Uses Firebase to store hours"
- Vague expectations: "Should be user-friendly"
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Example User Story:
As a [patient], I want [to control whether my medical images are used in training], so that [I can protect my privacy and feel safe about how my data is handled].

Example Acceptance Criteria:

- Given I am completing hospital consent forms, When I am asked about the use of my medical data for educational purposes, Then I must have a clear and separate option to either consent or decline.
- Given I have opted in, When my data is accessed by the training system, Then I must receive a notification in my patient portal that summarizes when and how it was used.

Write a few more acceptance criteria for the example user story:

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Finally, we can use the MoSCoW method to categorize and prioritize our acceptance criteria.

The **MoSCoW** Method is a simple way to prioritize requirements by organizing them into four categories: Must have, Should have, Could have, and Won't have. The name comes from the capitalized letters—M, S, C, and W—with the "o"s added just to make it pronounceable.

This method helps teams focus on what really matters. **Each person may have a different categorization.** The MoSCoW method is intended to kickstart thinking about prioritization, and there is no one size fits all answer.

- Must have items are critical. Without them, the product or feature simply doesn't work, doesn't deliver value, or is unsafe or illegal.
- Should have items are important, but not essential. The product is still usable without them, but they add significant value.
- Could have items are nice to have. They can improve the experience but aren't necessary for success.
- Won't have means we're explicitly choosing not to include this for now—either because it's out of scope, not needed yet, or not worth the cost at this time.

After developing acceptance criteria, we can categorize them using the MoSCoW method to determine which features to work on first.

Categorize the following example acceptance criteria from the same user story using the MoSCoW method as you see fit:

As a [patient], I want [to control whether my medical images are used in training], so that [I can protect my privacy and feel safe about how my data is handled].

	Must Have	Should Have	Could Have	Won't Have
Given I am completing hospital consent forms, When I am asked about the use of my medical data for educational purposes, Then I must have a clear and separate option to either consent or decline.	○	○	○	○
Given I have opted in, When my data is	○	○	○	○

	Must Have	Should Have	Could Have	Won't Have
accessed by the training system, Then I must receive a notification in my patient portal that summarizes when and how it was used.				
Given I previously gave consent to data use, When I change my preference, Then I must be able to revoke consent through the hospital's patient portal or by submitting a formal request.	○	○	○	○
Given I revoke consent, When the system checks eligibility for case inclusion, Then my data must be automatically excluded from any training-facing interface.	○	○	○	○
Given I am unsure about my current status, When I access the patient portal, Then I must see a clear and up-to-date record of my consent choice and a brief explanation of what that means.	○	○	○	○

Minimal Training

You will be receiving minimal Requirements Engineering training. Requirements Engineering is the process of identifying, documenting, and managing what a system should do. It is a key part of both systems and software engineering. In this training, you'll learn how to communicate with stakeholders or customers to better understand their needs and define the right software solution for their specific context.

Please do not google, look up any terms or examples, or use generative AI (e.g. ChatGPT, etc.). Just try your best. The goal of the survey is NOT to get the answers all correct, but to be exposed to the material.

I understand and agree:



A few short definitions will be given:

User Story:

A user story is a short, simple description of something a user wants to do, written from the user's perspective.

The general format is:

- **As a [type of user], I want [to do something], so that [I achieve this goal].**

Acceptance criteria:

Acceptance Criteria are clear, testable conditions that must be met for the story to be considered complete. Acceptance criteria can be written for each user story in the form of "Given," "When," Then."

The MoSCoW method:

The **MoSCoW** Method is a simple way to prioritize requirements by organizing them into four categories: Must have, Should have, Could have, and Won't have.

- Must have items are critical.
- Should have items are important, but not essential.
- Could have items are nice to have.
- Won't have means we're explicitly choosing not to include this for now.

Transition Block

In the next sections, you'll be shown 4 stakeholder scenarios.

To help with the tasks ahead, you'll be given a document titled **"AI System Components"**. You can download and refer to it at any point during the study. Please take a moment to quickly familiarize yourself with it.

This document breaks down an AI system into five key components:

- **Data Type** – What kind of data will the AI system take as input?
- **Learning Paradigm** – How will the AI system learn from that data?
- **Model Type/Technique** – What kind of model or method will be used to build the system?
- **System Behavior/Setting** – How will the AI system act or operate in the real world?
- **Training Regime** – How will the system be initially trained?

Your task will be to:

1. Read a new stakeholder scenario.
2. Write user stories, acceptance criteria, and prioritize features using the MoSCoW method.
3. Choose one option from each of the five component categories to propose an AI system solution tailored to the scenario.

You may download the documents for your reference at any time during the study:

DOWNLOAD AI SYSTEM COMPONENTS PDF HERE:

AI System Components

DO NOT DISTRIBUTE TO OTHER PARTICIPANTS.

Stakeholder 1

Hello! My name is Cady, and I work at our local emergency dispatch center. On a normal day, I help answer emergency calls, document information, prioritize which resources need to be sent out, and coordinate with first responders.

We get all sorts of calls, from minor injuries to people being trapped and unable to move, which all demand different levels of resources. During times of high call activity, it's hard to process incoming calls, prioritize them accurately, and dispatch the right resources as fast as we can. Because of this, my manager is thinking about how we can revamp our dispatch system to better handle these surges.

We've unfortunately had incidents where calls got delayed because we simply couldn't process them fast enough. In one case, there was a huge earthquake and a family trapped in their home didn't get immediate attention because their call got buried under dozens of minor injury reports. We tried our best to get through the queues, but we can never really predict what will happen.

Usually, people calling in around here are only speaking english. I was even wondering if we could use something like Siri or some technology that can already interpret English calls. Those exist, right? I guess they wouldn't really be able to recognize emergency calls though since they're not really like normal conversations most of the time. I also wouldn't want some Siri-thing to mistakenly say that some situation isn't serious when it is. For example, sometimes we get calls where someone is ordering a pizza, but they're really reporting a domestic abuse situation.

At least a consistent thing is that people typically report the same types of emergencies: broken bones, chest pain, house fires, gas leaks, people stuck in elevators. But every once in a while we get a weird or unexpected call, like a grown adult who got stuck in a kitchen chair. It's entertaining to look back at our logs and listen to the crazy calls.

There really is a lot of information to process during every call. First, someone has to figure out exactly what the caller is saying and type that information into the system. Then they have to determine how urgent the situation is and whether it is life-threatening or something that can wait. Finally, they have to decide what kinds of resources to allocate: ambulance, fire department, search and rescue, police, or some combination. The process of logging all this information is arduous, but as a result we have a lot of recordings of calls and records of what resources were deployed for each call.

I recently pitched an idea to our leadership team for a smart dispatch tool that could take over dispatch for the more routine calls and route the complicated ones to humans. Leadership thought this kind of tool might be useful, but they also raised some concerns. What if the system misses something important during one of those rare, weird cases? What if it doesn't understand the local quirks, like how certain backroads flood after an earthquake or how it's faster to take a side street to the hospital than the main road? Also, apparently our whole tech allocation for the year is only for a couple of regular office computers, so we can only rely on cloud-based services if we need something stronger.

Personally, I think we can't rely on a smart system for every case. But for the common calls like broken arms, smoke alarms, and minor injuries, I think automation could help us manage the call volume and let humans focus on the high-stakes situations where experience really matters.

Identify the **key stakeholders** for this scenario. How many distinct types of users do you think there are?

▶ 📌 **Click here to expand and see the scenario from Cady again** 📌

Write user stories for stakeholder 1. Each story should be on a separate line. Number each story.

Format: As a [type of user], **I** want [to do something], so that [**I** achieve this goal]. (you do not need to include the square brackets)

Example:

1. As a type of user, I want to do something, so that **I** achieve this goal.
2. As a type of user, I want to do something, so that **I** achieve this goal.
3. As a type of user, I want to do something, so that **I** achieve this goal.
4. etc.

Write acceptance criteria for each user story you created for stakeholder 1.

- Use the format: Given... when... then...
- You may write more than one criterion per user story.
- Label each criterion with the number of the user story it belongs to (e.g., 1., 2.,)

- Start each new criterion on a new line, even if a single criterion wraps onto multiple lines
- Put one criterion per line

Example Response:
1. Given I am completing hospital consent forms, when I am asked about the use of my medical data for educational purposes, then I must have a clear and separate option to either consent or decline**. (belongs to user story 1)**
1. Given I have opted in, when my data is accessed by the training system, then I must receive a notification in my patient portal that summarizes when and how it was used**. (belongs to user story 1)**
2. Given... when... then... **(belongs to user story 2)**
3. Given... when... then... **(belongs to user story 3)**

Label **each acceptance criteria for stakeholder 1** using the **MoSCoW method by adding one of the following tags at the beginning of each line.**

- Use **[M]** for Must
- Use **[S]** for Should
- Use **[C]** for Could
- Use **[W]** for Won't

You can **copy and paste your responses from the previous question** to get started.

Example:

1. [M] Given I am completing hospital consent forms, when I am asked about the use of my medical data for educational purposes, then I must have a clear and separate option to either consent or decline**. (belongs to user story 1)**

1. [S] Given I have opted in, when my data is accessed by the training system, then I must receive a notification in my patient portal that summarizes when and how it was used**. (belongs to user story 1)**

2. [S] Given... when... then... **(belongs to user story 2)**

3. [C] Given... when... then... **(belongs to user story 3)**

▶ 📌 **Click here to expand and see the scenario from Cady again** 📌

Write user stories for stakeholder 2. Each story should be on a **separate line. Number each story.**

Format: As a [type of user], I want [to do something], so that [I achieve this goal]**.** (you do not need to include the square brackets)

Example:

1. As a type of user, I want to do something, so that I achieve this goal.
2. As a type of user, I want to do something, so that I achieve this goal.
3. As a type of user, I want to do something, so that I achieve this goal.
4. etc.

Write acceptance criteria for each user story you created for stakeholder 2.

- Use the format: Given... when... then...
- You may write more than one criterion per user story.
- Label each criterion with the number of the user story it belongs to (e.g., 1., 2.,)
- Start each new criterion on a new line, even if a single criterion wraps onto multiple lines
- Put one criterion per line

Example Response:

1. Given I am completing hospital consent forms, when I am asked about the use of my medical data for educational purposes, then I must have a clear and separate option to either consent or decline**. (belongs to user story 1)**

1. Given I have opted in, when my data is accessed by the training system, then I must receive a notification in my patient portal that summarizes when and how it was used**. (belongs to user story 1)**

2. Given... when... then... **(belongs to user story 2)**

3. Given... when... then... **(belongs to user story 3)**

Label **each acceptance criteria for stakeholder 2** using the **MoSCoW method by adding one of the following tags at the beginning of each line.**

- Use **[M]** for Must
- Use **[S]** for Should
- Use **[C]** for Could
- Use **[W]** for Won't

You can **copy and paste your responses from the previous question** to get started.

Example:
1. [M] Given I am completing hospital consent forms, when I am asked about the use of my medical data for educational purposes, then I must have a clear and separate option to either consent or decline**. (belongs to user story 1)**
1. [S] Given I have opted in, when my data is accessed by the training system, then I must receive a notification in my patient portal that summarizes when and how it was used**. (belongs to user story 1)**
2. [S] Given... when... then... **(belongs to user story 2)**
3. [C] Given... when... then... **(belongs to user story 3)**

▶ 📌 **Click here to expand and see the scenario from Cady again** 📌

Write user stories for stakeholder 3. Each story should be on a **separate line. Number each story.**

Format: As a [type of user], I want [to do something], so that [I achieve this goal]. (you do not need to include the square brackets)

Example:

1. As a type of user, I want to do something, so that I achieve this goal.
2. As a type of user, I want to do something, so that I achieve this goal.
3. As a type of user, I want to do something, so that I achieve this goal.
4. etc.

Write acceptance criteria for each user story you created for stakeholder 3.

- Use the format: Given... when... then...
- You may write more than one criterion per user story.
- Label each criterion with the number of the user story it belongs to (e.g., 1., 2.,)
- Start each new criterion on a new line, even if a single criterion wraps onto multiple lines
- Put one criterion per line

Example Response:

1. Given I am completing hospital consent forms, when I am asked about the use of my medical data for educational purposes, then I must have a clear and separate option to either consent or decline**. (belongs to user story 1)**
1. Given I have opted in, when my data is accessed by the training system, then I must receive a notification in my patient portal that summarizes when and how it was used**. (belongs to user story 1)**
2. Given... when... then... **(belongs to user story 2)**
3. Given... when... then... **(belongs to user story 3)**

Label **each acceptance criteria for stakeholder 3** using the **MoSCoW method by adding one of the following tags at the beginning of each line.**

- Use **[M]** for Must
- Use **[S]** for Should
- Use **[C]** for Could
- Use **[W]** for Won't

You can **copy and paste your responses from the previous question** to get started.

Example:
1. [M] Given I am completing hospital consent forms, when I am asked about the use of my medical data for educational purposes, then I must have a clear and separate option to either consent or decline**. (belongs to user story 1)**
1. [S] Given I have opted in, when my data is accessed by the training system, then I must receive a notification in my patient portal that summarizes when and how it was used**. (belongs to user story 1)**

2. [S] Given... when... then... **(belongs to user story 2)**
3. [C] Given... when... then... **(belongs to user story 3)**

▶ 📌 **Click here to expand and see the scenario from Cady again** 📌

Write user stories for stakeholder 4. Each story should be on a **separate line. Number each story.**

Format: As a [type of user], **I** want [to do something], so that [**I** achieve this goal]**.** (you do not need to include the square brackets)

Example:

1. As a type of user, I want to do something, so that **I** achieve this goal.
2. As a type of user, I want to do something, so that **I** achieve this goal.
3. As a type of user, I want to do something, so that **I** achieve this goal.
4. etc.

Write acceptance criteria for each user story you created for stakeholder 4.

- Use the format: Given... when... then...
- You may write more than one criterion per user story.
- Label each criterion with the number of the user story it belongs to (e.g., 1., 2.,)
- Start each new criterion on a new line, even if a single criterion wraps onto multiple lines
- Put one criterion per line

Example Response:
1. Given I am completing hospital consent forms, when I am asked about the use of my medical data for educational purposes, then I must have a clear and separate option to either consent or decline**. (belongs to user story 1)**
1. Given I have opted in, when my data is accessed by the training system, then I must receive a notification in my patient portal that summarizes when and how it was used**. (belongs to user story 1)**
2. Given... when... then... **(belongs to user story 2)**
3. Given... when... then... **(belongs to user story 3)**

Label **each acceptance criteria for stakeholder 4** using the **MoSCoW method by adding one of the following tags at the beginning of each line.**

- Use **[M]** for Must
- Use **[S]** for Should
- Use **[C]** for Could
- Use **[W]** for Won't

You can **copy and paste your responses from the previous question** to get started.

Example:
1. [M] Given I am completing hospital consent forms, when **I** am asked about the use of my medical data for educational purposes, then I must have a clear and separate option to either consent or decline**. (belongs to user story 1)**
1. [S] Given **I** have opted in, when my data is accessed by the training system, then I must receive a notification in my patient portal that summarizes when and how it was used**. (belongs to user story 1)**
2. [S] Given... when... then... **(belongs to user story 2)**
3. [C] Given... when... then... **(belongs to user story 3)**

▶ 📌 **Click here to expand and see the scenario from Cady again** 📌

Write user stories for stakeholder 5. Each story should be on a **separate line. Number each story.**

Format: As a [type of user], **I** want [to do something], so that [**I** achieve this goal]**.** (you do not need to include the square brackets)

Example:

1. As a type of user, I want to do something, so that **I** achieve this goal.

2. As a type of user, I want to do something, so that I achieve this goal.
3. As a type of user, I want to do something, so that I achieve this goal.
4. etc.

Write acceptance criteria for each user story you created for stakeholder 5.

- Use the format: Given... when... then...
- You may write more than one criterion per user story.
- Label each criterion with the number of the user story it belongs to (e.g., 1., 2.,)
- Start each new criterion on a new line, even if a single criterion wraps onto multiple lines
- Put one criterion per line

Example Response:

1. Given I am completing hospital consent forms, when I am asked about the use of my medical data for educational purposes, then I must have a clear and separate option to either consent or decline**. (belongs to user story 1)**

1. Given I have opted in, when my data is accessed by the training system, then I must receive a notification in my patient portal that summarizes when and how it was used**. (belongs to user story 1)**

2. Given... when... then... **(belongs to user story 2)**

3. Given... when... then... **(belongs to user story 3)**

Label **each acceptance criteria for stakeholder 5** using the **MoSCoW method by adding one of the following tags at the beginning of each line.**

- Use **[M]** for Must
- Use **[S]** for Should
- Use **[C]** for Could
- Use **[W]** for Won't

You can **copy and paste your responses from the previous question** to get started.

Example:
1. [M] Given I am completing hospital consent forms, when I am asked about the use of my medical data for educational purposes, then I must have a clear and separate option to either consent or decline**. (belongs to user story 1)**
1. [S] Given I have opted in, when my data is accessed by the training system, then I must receive a notification in my patient portal that summarizes when and how it was used**. (belongs to user story 1)**
2. [S] Given... when... then... **(belongs to user story 2)**
3. [C] Given... when... then... **(belongs to user story 3)**

▶ 📌 **Click here to expand and see the scenario from Cady again** 📌

Based on the information you have from this scenario, imagine what the best AI system solution would be after an iterative development process. What kind of DATA TYPE would this system use?

Use the DATA TYPE section in the PDF to guide your decision.

- ○ Numeric
- ○ Tabular
- ○ Images
- ○ Audio
- ○ Time Series

Based on the information you have from this scenario, imagine what the best AI system solution would be after an iterative development process. What kind of LEARNING PARADIGM would this system use?

Use the LEARNING PARADIGM section in the PDF to guide your decision.

- ○ Supervised Learning
- ○ Unsupervised Learning
- ○ Reinforcement Learning
- ○ Semi-Supervised Learning
- ○ Imitation Learning
- ○ Human-in-the-loop Learning

Based on the information you have from this scenario, imagine what the best AI system solution would be after an iterative development process. What kind of MODEL TYPE/TECHNIQUE would this system use?

Use the MODEL TYPE/TECHNIQUES section in the PDF to guide your decision.

- ○ Graph Neural Networks
- ○ Transformers
- ○ Bandits
- ○ Clustering
- ○ Rule-Based
- ○ Diffusion Models

Based on the information you have from this scenario, imagine what the best AI system solution would be after an iterative development process. What kind of SYSTEM BEHAVIOR/SETTING would this system have?

Use the SYSTEM BEHAVIOR/SETTING section in the PDF to guide your decision.

- ○ Complementary Systems
- ○ Cooperative Systems
- ○ Fully Interactive
- ○ Decision Support
- ○ Autonomous

Based on the information you have from this scenario, imagine what the best AI system solution would be after an iterative development process. What kind of TRAINING REGIME would this system use?

Use the TRAINING REGIME section in the PDF to guide your decision.

- ○ Transfer Learning
- ○ Few-Shot/Meta Learning
- ○ Pre-Trained
- ○ Trained from Scratch

Stakeholder 2

[redacted]

I'm Josie, and I work for NRG. We manage the energy and power stations around this side of the state. Most days, I'm out in the field checking on substations, monitoring power lines, and honestly, just making sure nothing's on fire. Literally and figuratively.

But lately, man. The weather's been crazy, and it's been throwing everything off. When it gets hot, the grid gets seriously stressed. Some areas suck up a ton of electricity trying to keep AC units blasting, while others just can't get enough. We've even had reports of elderly residents getting heatstroke because their homes can't stay cool. My manager's been pushing hard for us to fix it.

The town next to us actually upgraded to this really slick smart grid setup. It can shift power dynamically based on who needs it most. Our residents deserve the same thing, and we've been trying to convince the higher-ups to invest in new infrastructure so we can either join their grid or build a smart system of our own.

I'm not gonna lie…I don't totally get the technical side. There's something about a fancy algorithm one of their engineers cooked up for their grid. My manager's got some contacts over there, so maybe we can snag their secret sauce and use it as a starting point. And if I remember correctly, we do have some beefy computers that we can work with, but not much budget for anything else.

I think there's a catch, though. Their town's full of shiny new buildings that are pretty easy to work with. Ours is a patchwork of half old, half brand-new developments. That means whatever system we use might need some serious tweaks to actually work here. Also, we don't even have a perfect map of where the old stuff and new stuff is. No one's bothered to lay it out properly, and Google Maps around here is slow to catch up, even if the buildings are already wired into the grid and pulling power.

The good news is, we already collect tons of data: power loads, substation temperatures, wind patterns, weather forecasts, all time-stamped and updated every minute, which is pretty great. But man, one thing I wish we had is better visibility between substations. Right now, we can see what's happening at each station, but not how the power's doing on the lines between them.

That means even if we think we're sending enough power somewhere, houses stuck

between two substations in our “blind spots” might not actually be getting what they need. We only find out if they call to complain, or if they don’t, which we just have to assume means they’re fine. And even then, every resident’s different. Some people don’t mind a short outage now and then. Others absolutely need steady, reliable power 24/7. Luckily, we’ve kept a list of complaints so we know who to give some extra attention to already.

The dream is to have a system smart enough to read all the data, predict what’s coming, and balance power on its own so we’re not stuck playing whack-a-mole every time the weather changes or everyone cranks their AC. I think the most important thing is probably that our system is accurate to our town. Since we won’t get regular severe weather for months, we have some time to spare before we want to get something up and running.

Identify the **key stakeholders** for this scenario. How many distinct types of users do you think there are?

▶ 📌 **Click here to expand and see the scenario from Josie again** 📌

Write user stories for stakeholder 1. Each story should be on a separate line. Number each story.

Format: As a [type of user], I want [to do something], so that [I achieve this goal]. (you do not need to include the square brackets)

Example:

1. As a type of user, I want to do something, so that I achieve this goal.
2. As a type of user, I want to do something, so that I achieve this goal.
3. As a type of user, I want to do something, so that I achieve this goal.
4. etc.

Write acceptance criteria for each user story you created for stakeholder 1.

- Use the format: Given... when... then...
- You may write more than one criterion per user story.
- Label each criterion with the number of the user story it belongs to (e.g., 1., 2.,)
- Start each new criterion on a new line, even if a single criterion wraps onto multiple lines
- Put one criterion per line

Example Response:

1. Given I am completing hospital consent forms, when I am asked about the use of my medical data for educational purposes, then I must have a clear and separate option to either consent or decline**. (belongs to user story 1)**

1. Given I have opted in, when my data is accessed by the training system, then I must receive a notification in my patient portal that summarizes when and how it was used**. (belongs to user story 1)**

2. Given... when... then... **(belongs to user story 2)**

3. Given... when... then... **(belongs to user story 3)**

Label **each acceptance criteria for stakeholder 1** using the **MoSCoW method by adding one of the following tags at the beginning of each line.**

- Use **[M]** for Must
- Use **[S]** for Should
- Use **[C]** for Could
- Use **[W]** for Won't

You can **copy and paste your responses from the previous question** to get started.

Example:
1. [M] Given I am completing hospital consent forms, when I am asked about the use of my medical data for educational purposes, then I must have a clear and separate option to either consent or decline**. (belongs to user story 1)**
1. [S] Given I have opted in, when my data is accessed by the training system, then I must receive a notification in my patient portal that summarizes when and how it was used**. (belongs to user story 1)**
2. [S] Given... when... then... **(belongs to user story 2)**
3. [C] Given... when... then... **(belongs to user story 3)**

▶ 📌 **Click here to expand and see the scenario from Josie again** 📌

Write user stories for stakeholder 2. Each story should be on a **separate line. Number each story.**

Format: As a [type of user], **I** want [to do something], so that [**I** achieve this goal]**.** (you do not need to include the square brackets)

Example:

1. As a type of user, I want to do something, so that **I** achieve this goal.
2. As a type of user, I want to do something, so that **I** achieve this goal.
3. As a type of user, I want to do something, so that **I** achieve this goal.
4. etc.

Write acceptance criteria for each user story you created for stakeholder 2.

- Use the format: Given... when... then...
- You may write more than one criterion per user story.
- Label each criterion with the number of the user story it belongs to (e.g., 1., 2.,)
- Start each new criterion on a new line, even if a single criterion wraps onto multiple lines
- Put one criterion per line

Example Response:

1. Given I am completing hospital consent forms, when I am asked about the use of my medical data for educational purposes, then I must have a clear and separate option to either consent or decline. **(belongs to user story 1)**

1. Given I have opted in, when my data is accessed by the training system, then I must receive a notification in my patient portal that summarizes when and how it was used. **(belongs to user story 1)**

2. Given... when... then... **(belongs to user story 2)**

3. Given... when... then... **(belongs to user story 3)**

Label **each acceptance criteria for stakeholder 2** using the **MoSCoW method by adding one of the following tags at the beginning of each line.**

- Use **[M]** for Must
- Use **[S]** for Should
- Use **[C]** for Could
- Use **[W]** for Won't

You can **copy and paste your responses from the previous question** to get started.

Example:

1. [M] Given I am completing hospital consent forms, when I am asked about the use of my medical data for educational purposes, then I must have a clear and separate option to either consent or decline. **(belongs to user story 1)**

1. [S] Given I have opted in, when my data is accessed by the training system, then I

must receive a notification in my patient portal that summarizes when and how it was used. **(belongs to user story 1)**

2. [S] Given... when... then... **(belongs to user story 2)**

3. [C] Given... when... then... **(belongs to user story 3)**

▶ 📌 **Click here to expand and see the scenario from Josie again** 📌

Write user stories for stakeholder 3. Each story should be on a **separate line. Number each story.**

Format: As a [type of user], **I** want [to do something], so that [**I** achieve this goal]**.** (you do not need to include the square brackets)

Example:

1. As a type of user, I want to do something, so that **I** achieve this goal.
2. As a type of user, I want to do something, so that **I** achieve this goal.
3. As a type of user, I want to do something, so that **I** achieve this goal.
4. etc.

Write acceptance criteria for each user story you created for stakeholder 3.

- Use the format: Given... when... then...
- You may write more than one criterion per user story.
- Label each criterion with the number of the user story it belongs to (e.g., 1., 2.,)
- Start each new criterion on a new line, even if a single criterion wraps onto multiple lines
- Put one criterion per line

Example Response:
1. Given I am completing hospital consent forms, when I am asked about the use of my medical data for educational purposes, then I must have a clear and separate option to either consent or decline**. (belongs to user story 1)**
1. Given I have opted in, when my data is accessed by the training system, then I must receive a notification in my patient portal that summarizes when and how it was used**. (belongs to user story 1)**
2. Given... when... then... **(belongs to user story 2)**
3. Given... when... then... **(belongs to user story 3)**

Label **each acceptance criteria for stakeholder 3** using the **MoSCoW method by adding one of the following tags at the beginning of each line.**

- Use **[M]** for Must
- Use **[S]** for Should
- Use **[C]** for Could

- Use **[W]** for Won't

You can **copy and paste your responses from the previous question** to get started.

Example:
1. [M] Given I am completing hospital consent forms, when I am asked about the use of my medical data for educational purposes, then I must have a clear and separate option to either consent or decline**. (belongs to user story 1)**
1. [S] Given I have opted in, when my data is accessed by the training system, then I must receive a notification in my patient portal that summarizes when and how it was used**. (belongs to user story 1)**
2. [S] Given... when... then... **(belongs to user story 2)**
3. [C] Given... when... then... **(belongs to user story 3)**

▶ 📌 **Click here to expand and see the scenario from Josie again** 📌

Write user stories for stakeholder 4. Each story should be on a **separate line. Number each story.**

Format: As a [type of user], I want [to do something], so that [I achieve this goal]. (you do not need to include the square brackets)

Example:

1. As a type of user, I want to do something, so that I achieve this goal.
2. As a type of user, I want to do something, so that I achieve this goal.

3. As a type of user, I want to do something, so that I achieve this goal.
4. etc.

Write acceptance criteria for each user story you created for stakeholder 4.

- Use the format: Given... when... then...
- You may write more than one criterion per user story.
- Label each criterion with the number of the user story it belongs to (e.g., 1., 2.,)
- Start each new criterion on a new line, even if a single criterion wraps onto multiple lines
- Put one criterion per line

Example Response:

1. Given I am completing hospital consent forms, when I am asked about the use of my medical data for educational purposes, then I must have a clear and separate option to either consent or decline. **(belongs to user story 1)**

1. Given I have opted in, when my data is accessed by the training system, then I must receive a notification in my patient portal that summarizes when and how it was used. **(belongs to user story 1)**

2. Given... when... then... **(belongs to user story 2)**

3. Given... when... then... **(belongs to user story 3)**

Label **each acceptance criteria for stakeholder 4** using the **MoSCoW method by adding one of the following tags at the beginning of each line.**

- Use **[M]** for Must
- Use **[S]** for Should
- Use **[C]** for Could
- Use **[W]** for Won't

You can **copy and paste your responses from the previous question** to get started.

Example:
1. [M] Given I am completing hospital consent forms, when I am asked about the use of my medical data for educational purposes, then I must have a clear and separate option to either consent or decline**. (belongs to user story 1)**
1. [S] Given I have opted in, when my data is accessed by the training system, then I must receive a notification in my patient portal that summarizes when and how it was used**. (belongs to user story 1)**
2. [S] Given... when... then... **(belongs to user story 2)**
3. [C] Given... when... then... **(belongs to user story 3)**

▶ 📌 **Click here to expand and see the scenario from Josie again** 📌

Write user stories for stakeholder 5. Each story should be on a **separate line. Number each story.**

Format: As a [type of user], I want [to do something], so that [I achieve this goal]. (you do not need to include the square brackets)

Example:

1. As a type of user, I want to do something, so that I achieve this goal.
2. As a type of user, I want to do something, so that I achieve this goal.
3. As a type of user, I want to do something, so that I achieve this goal.
4. etc.

Write acceptance criteria for each user story you created for stakeholder 5.

- Use the format: Given... when... then...
- You may write more than one criterion per user story.
- Label each criterion with the number of the user story it belongs to (e.g., 1., 2.,)
- Start each new criterion on a new line, even if a single criterion wraps onto multiple lines
- Put one criterion per line

Example Response:

1. Given I am completing hospital consent forms, when **I** am asked about the use of my medical data for educational purposes, then **I** must have a clear and separate option to either consent or decline**. (belongs to user story 1)**

1. Given I have opted in, when my data is accessed by the training system, then **I** must receive a notification in my patient portal that summarizes when and how it was used**. (belongs to user story 1)**

2. Given... when... then... **(belongs to user story 2)**

3. Given... when... then... **(belongs to user story 3)**

Label **each acceptance criteria for stakeholder 5** using the **MoSCoW method by adding one of the following tags at the beginning of each line.**

- Use **[M]** for Must
- Use **[S]** for Should
- Use **[C]** for Could
- Use **[W]** for Won't

You can **copy and paste your responses from the previous question** to get started.

Example:

1. [M] Given I am completing hospital consent forms, when **I** am asked about the use of my medical data for educational purposes, then I must have a clear and separate option to either consent or decline**. (belongs to user story 1)**

1. [S] Given **I** have opted in, when my data is accessed by the training system, then I must receive a notification in my patient portal that summarizes when and how it was

used. **(belongs to user story 1)**

2. [S] Given... when... then... **(belongs to user story 2)**

3. [C] Given... when... then... **(belongs to user story 3)**

▶ 📌 **Click here to expand and see the scenario from Josie again** 📌

Based on the information you have from this scenario, imagine what the best AI system solution would be after an iterative development process. What kind of DATA TYPE would this system use?

Use the DATA TYPE section in the PDF to guide your decision.

- ○ Numeric
- ○ Tabular
- ○ Images
- ○ Audio
- ○ Time Series

Based on the information you have from this scenario, imagine what the best AI system solution would be after an iterative development process. What kind of LEARNING PARADIGM would this system use?

Use the LEARNING PARADIGM section in the PDF to guide your decision.

- ○ Supervised Learning
- ○ Unsupervised Learning

- ◯ Reinforcement Learning
- ◯ Semi-Supervised Learning
- ◯ Imitation Learning
- ◯ Human-in-the-loop Learning

Based on the information you have from this scenario, imagine what the best AI system solution would be after an iterative development process. What kind of MODEL TYPE/TECHNIQUE would this system use?

Use the MODEL TYPE/TECHNIQUES section in the PDF to guide your decision.

- ◯ Graph Neural Networks
- ◯ Transformers
- ◯ Bandits
- ◯ Clustering
- ◯ Rule-Based
- ◯ Diffusion Models

Based on the information you have from this scenario, imagine what the best AI system solution would be after an iterative development process. What kind of SYSTEM BEHAVIOR/SETTING would this system have?

Use the SYSTEM BEHAVIOR/SETTING section in the PDF to guide your decision.

- ◯ Complementary Systems
- ◯ Cooperative Systems
- ◯ Fully Interactive Systems
- ◯ Decision Support Systems
- ◯ Autonomous

Based on the information you have from this scenario, imagine what the best AI system solution would be after an iterative development process. What kind of TRAINING REGIME would this system use?

Use the TRAINING REGIME section in the PDF to guide your decision.

- ◯ Transfer Learning
- ◯ Few-Shot/Meta Learning
- ◯ Pre-Trained
- ◯ Trained from Scratch

Stakeholder 3

Hi, my name is Ben. I work for the local government, in the branch that handles public assistance programs.

Our team is responsible for reviewing applications from people who are seeking things like housing support, food assistance, or medical subsidies. We have clear guidelines for who qualifies, such as income limits, residency status, and family size. But the reality is that it’s not always black and white.

Sometimes applicants' situations don’t fit neatly into a checkbox. They might have informal income sources, missing documents, or complicated living arrangements. Figuring out if someone truly meets the eligibility criteria requires a lot of effort. Form field data in our application often leaves out the content that truly exposes their situation. The best fix would be a full overhaul of the application system, but given its age and technical complexity, our only practical option is to build on top of it. Luckily, our engineers regularly receive external training on the latest software and AI methods, so they have the expertise to enhance and build on top of these older systems.

We get a large volume of applications every month, and our team struggles to keep up. We want to be fair and thorough, but the caseload makes it hard. Some people who urgently need help might end up waiting longer than they should. That means that

mothers who need support with getting baby formula have to wait, or someone experiencing homelessness might not get access to resources for weeks or months.

We really need to reduce our workload. I'm reading so many applications that as the day goes on, I'm so fatigued that I just can't pick out the finer details from applications anymore. We're hoping to get some sort of automated system that could help our staff by reviewing applications and putting them into buckets of "likely eligible, unclear, or likely ineligible." That way, we could quickly review their information, accept or reject the clear ones, and follow up with individuals to collect additional details for the ambiguous cases. I'm not sure if it should give us the answers directly, because I'm scared that if it makes a mistake then it would reflect poorly on us workers.

We know it wouldn't be perfect, which is why we'd keep our staff involved. This kind of categorization is really nuanced, which is why we have such experienced workers handling these cases. The system might not get everything right at first, but if we could flag correct or incorrect suggestions, maybe it could learn over time.

If we can build something like this, it would help us serve our community faster and more fairly. It would make sure the right help reaches the people who need it most, without overwhelming the people who work so hard to process every case.

Identify the **key stakeholders** for this scenario. How many distinct types of users do you think there are?

▶ 📌 **Click here to expand and see the scenario from Ben again** 📌

Write user stories for stakeholder 1. Each story should be on a separate line. Number each story.

Format: As a [type of user], I want [to do something], so that [I achieve this goal]. (you do not need to include the square brackets)

Example:

1. As a type of user, I want to do something, so that I achieve this goal.
2. As a type of user, I want to do something, so that I achieve this goal.
3. As a type of user, I want to do something, so that I achieve this goal.
4. etc.

Write acceptance criteria for each user story you created for stakeholder 1.

- Use the format: Given... when... then...
- You may write more than one criterion per user story.
- Label each criterion with the number of the user story it belongs to (e.g., 1., 2.,)
- Start each new criterion on a new line, even if a single criterion wraps onto multiple lines
- Put one criterion per line

Example Response:

1. Given I am completing hospital consent forms, when I am asked about the use of my medical data for educational purposes, then I must have a clear and separate option to either consent or decline**. (belongs to user story 1)**

1. Given I have opted in, when my data is accessed by the training system, then I must receive a notification in my patient portal that summarizes when and how it was used**. (belongs to user story 1)**

2. Given... when... then... **(belongs to user story 2)**

3. Given... when... then... **(belongs to user story 3)**

Label **each acceptance criteria for stakeholder 1** using the **MoSCoW method by adding one of the following tags at the beginning of each line.**

- Use **[M]** for Must
- Use **[S]** for Should
- Use **[C]** for Could
- Use **[W]** for Won't

You can **copy and paste your responses from the previous question** to get started.

Example:
1. [M] Given I am completing hospital consent forms, when I am asked about the use of my medical data for educational purposes, then I must have a clear and separate option to either consent or decline. **(belongs to user story 1)**
1. [S] Given I have opted in, when my data is accessed by the training system, then I must receive a notification in my patient portal that summarizes when and how it was used**. (belongs to user story 1)**
2. [S] Given... when... then... **(belongs to user story 2)**
3. [C] Given... when... then... **(belongs to user story 3)**

▶ 📌 **Click here to expand and see the scenario from Ben again** 📌

Write user stories for stakeholder 2. Each story should be on a **separate line. Number each story.**

Format: As a [type of user], **I** want [to do something], so that [**I** achieve this goal]**.** (you do not need to include the square brackets)

Example:

1. As a type of user, I want to do something, so that **I** achieve this goal.
2. As a type of user, I want to do something, so that **I** achieve this goal.
3. As a type of user, I want to do something, so that **I** achieve this goal.
4. etc.

Write acceptance criteria for each user story you created for stakeholder 2.

- Use the format: Given... when... then...
- You may write more than one criterion per user story.

- Label each criterion with the number of the user story it belongs to (e.g., 1., 2.,)
- Start each new criterion on a new line, even if a single criterion wraps onto multiple lines
- Put one criterion per line

Example Response:
1. Given I am completing hospital consent forms, when I am asked about the use of my medical data for educational purposes, then I must have a clear and separate option to either consent or decline**. (belongs to user story 1)**
1. Given I have opted in, when my data is accessed by the training system, then I must receive a notification in my patient portal that summarizes when and how it was used**. (belongs to user story 1)**
2. Given... when... then... **(belongs to user story 2)**
3. Given... when... then... **(belongs to user story 3)**

Label **each acceptance criteria for stakeholder 2** using the **MoSCoW method by adding one of the following tags at the beginning of each line.**

- Use **[M]** for Must
- Use **[S]** for Should
- Use **[C]** for Could
- Use **[W]** for Won't

You can **copy and paste your responses from the previous question** to get started.

Example:

1. [M] Given I am completing hospital consent forms, when I am asked about the use of my medical data for educational purposes, then I must have a clear and separate option to either consent or decline. **(belongs to user story 1)**

1. [S] Given I have opted in, when my data is accessed by the training system, then I must receive a notification in my patient portal that summarizes when and how it was used. **(belongs to user story 1)**

2. [S] Given... when... then... **(belongs to user story 2)**

3. [C] Given... when... then... **(belongs to user story 3)**

▶ 📌 **Click here to expand and see the scenario from Ben again** 📌

Write user stories for stakeholder 3. Each story should be on a **separate line. Number each story.**

Format: As a [type of user], I want [to do something], so that [I achieve this goal]. (you do not need to include the square brackets)

Example:

1. As a type of user, I want to do something, so that I achieve this goal.
2. As a type of user, I want to do something, so that I achieve this goal.
3. As a type of user, I want to do something, so that I achieve this goal.
4. etc.

Write acceptance criteria for each user story you created for stakeholder 3.

- Use the format: Given... when... then...
- You may write more than one criterion per user story.
- Label each criterion with the number of the user story it belongs to (e.g., 1., 2.,)
- Start each new criterion on a new line, even if a single criterion wraps onto multiple lines
- Put one criterion per line

Example Response:

1. Given I am completing hospital consent forms, when I am asked about the use of my medical data for educational purposes, then I must have a clear and separate option to either consent or decline**. (belongs to user story 1)**

1. Given I have opted in, when my data is accessed by the training system, then I must receive a notification in my patient portal that summarizes when and how it was used. **(belongs to user story 1)**

2. Given... when... then... **(belongs to user story 2)**

3. Given... when... then... **(belongs to user story 3)**

Label **each acceptance criteria for stakeholder 3** using the **MoSCoW method by adding one of the following tags at the beginning of each line.**

- Use **[M]** for Must
- Use **[S]** for Should
- Use **[C]** for Could
- Use **[W]** for Won't

You can **copy and paste your responses from the previous question** to get started.

Example:
1. [M] Given I am completing hospital consent forms, when I am asked about the use of my medical data for educational purposes, then I must have a clear and separate option to either consent or decline**. (belongs to user story 1)**
1. [S] Given I have opted in, when my data is accessed by the training system, then I must receive a notification in my patient portal that summarizes when and how it was used**. (belongs to user story 1)**
2. [S] Given... when... then... **(belongs to user story 2)**
3. [C] Given... when... then... **(belongs to user story 3)**

▶ 📌 **Click here to expand and see the scenario from Ben again** 📌

Write user stories for stakeholder 4. Each story should be on a **separate line. Number each story.**

Format: As a [type of user], I want [to do something], so that [I achieve this goal]. (you do not need to include the square brackets)

Example:

1. As a type of user, I want to do something, so that I achieve this goal.
2. As a type of user, I want to do something, so that I achieve this goal.
3. As a type of user, I want to do something, so that I achieve this goal.
4. etc.

Write acceptance criteria for each user story you created for stakeholder 4.

- Use the format: Given... when... then...
- You may write more than one criterion per user story.
- Label each criterion with the number of the user story it belongs to (e.g., 1., 2.,)
- Start each new criterion on a new line, even if a single criterion wraps onto multiple lines
- Put one criterion per line

Example Response:

1. Given I am completing hospital consent forms, when I am asked about the use of my medical data for educational purposes, then I must have a clear and separate option to either consent or decline**. (belongs to user story 1)**

1. Given I have opted in, when my data is accessed by the training system, then I must receive a notification in my patient portal that summarizes when and how it was used**. (belongs to user story 1)**

2. Given... when... then... **(belongs to user story 2)**
3. Given... when... then... **(belongs to user story 3)**

Label **each acceptance criteria for stakeholder 4** using the **MoSCoW method by adding one of the following tags at the beginning of each line.**

- Use **[M]** for Must
- Use **[S]** for Should
- Use **[C]** for Could
- Use **[W]** for Won't

You can **copy and paste your responses from the previous question** to get started.

Example:
1. [M] Given I am completing hospital consent forms, when I am asked about the use of my medical data for educational purposes, then I must have a clear and separate option to either consent or decline**. (belongs to user story 1)**
1. [S] Given I have opted in, when my data is accessed by the training system, then I must receive a notification in my patient portal that summarizes when and how it was used**. (belongs to user story 1)**
2. [S] Given... when... then... **(belongs to user story 2)**
3. [C] Given... when... then... **(belongs to user story 3)**

▶ 📌 **Click here to expand and see the scenario from Ben again** 📌

Write user stories for stakeholder 5. Each story should be on a **separate line. Number each story.**

Format: As a [type of user], **I** want [to do something], so that [**I** achieve this goal]**.** (you do not need to include the square brackets)

Example:

1. As a type of user, I want to do something, so that **I** achieve this goal.
2. As a type of user, I want to do something, so that **I** achieve this goal.
3. As a type of user, I want to do something, so that **I** achieve this goal.
4. etc.

Write acceptance criteria for each user story you created for stakeholder 5.

- Use the format: Given... when... then...
- You may write more than one criterion per user story.
- Label each criterion with the number of the user story it belongs to (e.g., 1., 2.,)

- Start each new criterion on a new line, even if a single criterion wraps onto multiple lines
- Put one criterion per line

Example Response:
1. Given I am completing hospital consent forms, when I am asked about the use of my medical data for educational purposes, then I must have a clear and separate option to either consent or decline**. (belongs to user story 1)**
1. Given I have opted in, when my data is accessed by the training system, then I must receive a notification in my patient portal that summarizes when and how it was used**. (belongs to user story 1)**
2. Given... when... then... **(belongs to user story 2)**
3. Given... when... then... **(belongs to user story 3)**

Label **each acceptance criteria for stakeholder 5** using the **MoSCoW method by adding one of the following tags at the beginning of each line.**

- Use **[M]** for Must
- Use **[S]** for Should
- Use **[C]** for Could
- Use **[W]** for Won't

You can **copy and paste your responses from the previous question** to get started.

Example:

1. [M] Given I am completing hospital consent forms, when I am asked about the use of my medical data for educational purposes, then I must have a clear and separate option to either consent or decline**. (belongs to user story 1)**

1. [S] Given I have opted in, when my data is accessed by the training system, then I must receive a notification in my patient portal that summarizes when and how it was used**. (belongs to user story 1)**

2. [S] Given... when... then... **(belongs to user story 2)**

3. [C] Given... when... then... **(belongs to user story 3)**

▶ 📌 **Click here to expand and see the scenario from Ben again** 📌

Based on the information you have from this scenario, imagine what the best AI system solution would be after an iterative development process. What kind of DATA TYPE would this system use?

Use the DATA TYPE section in the PDF to guide your decision.

- ○ Numeric
- ○ Tabular
- ○ Images
- ○ Audio
- ○ Time Series

Based on the information you have from this scenario, imagine what the best AI system solution would be after an iterative development process. What kind of LEARNING

PARADIGM would this system use?

Use the LEARNING PARADIGM section in the PDF to guide your decision.

- ◯ Supervised Learning
- ◯ Unsupervised Learning
- ◯ Reinforcement Learning
- ◯ Semi-Supervised Learning
- ◯ Imitation Learning
- ◯ Human-in-the-loop Learning

Based on the information you have from this scenario, imagine what the best AI system solution would be after an iterative development process. What kind of MODEL TYPE/TECHNIQUE would this system use?

Use the MODEL TYPE/TECHNIQUES section in the PDF to guide your decision.

- ◯ Graph Neural Networks
- ◯ Transformers
- ◯ Bandits
- ◯ Clustering
- ◯ Rule-Based
- ◯ Diffusion Models

Based on the information you have from this scenario, imagine what the best AI system solution would be after an iterative development process. What kind of SYSTEM BEHAVIOR/SETTING would this system have?

Use the SYSTEM BEHAVIOR/SETTING section in the PDF to guide your decision.

- ◯ Complementary Systems
- ◯ Cooperative Systems
- ◯ Fully Interactive Systems

- ○ Decision Support Systems
- ○ Autonomous Systems

Based on the information you have from this scenario, imagine what the best AI system solution would be after an iterative development process. What kind of TRAINING REGIME would this system use?

Use the TRAINING REGIME section in the PDF to guide your decision.

- ○ Transfer Learning
- ○ Few-Shot/Meta Learning
- ○ Pre-Trained
- ○ Trained from Scratch

Stakeholder 4

Hi, my name is Megan. I work at a public high school in a lower-income urban area. Our school serves a very diverse population: students come from different racial backgrounds, immigrant families, foster care, single-parent households, and more. Over the years I've learned that a lot of our students have responsibilities outside of school, like working part-time, caring for younger siblings, or dealing with unstable living situations.

I have a lot of experience counseling students, but my caseload is over 300 students per semester. There's no way I can give every student the time and attention they deserve individually, but I can't just focus on a few groups either… Everyone deserves support.

Most days, I meet with students briefly in one on one meetings for 15 to 20 minutes. We go over their grades, attendance records, and talk about their career or college plans. My main goal is to spot problems early, while there's still time to intervene and offer help before small issues turn into big things we can't fix.

Some of the biggest challenges I face are the hidden problems. Students often don't tell

me everything upfront, which I understand! It can be hard to open up and be vulnerable. But sometimes, I only find out that a student was homeless after they've already dropped out. And unless there's something obvious, like a student failing all their classes, it's really hard to know who needs urgent help. It's also difficult to spot patterns across students. For example, if a whole group starts disengaging because an entire block lost internet access, I might not realize it until it's too late.

I have access to individual student files, but we don't have clear labels like "at-risk" or "fine." I can see obvious things like grades and absences, but most of the real risk factors — the things happening outside of school — aren't visible right away.

Our administration thought that part of the problem might be that students are too embarrassed to share their struggles directly with counselors. So they sent out an anonymous survey asking students about their living situations. The questions were things like, "Do you have reliable transportation to school?" and "Are you the full-time caretaker for another family member?" I think the admin plans to send out these surveys every few months to make sure that we stay relatively up to date without overwhelming the kids. We have some of these survey responses now, but it's so much data that it's been overwhelming to sort through and make sense of it.

What I really want is to figure out what the biggest struggles are for our students, and more importantly, how those struggles affect their academic performance. I want to understand what's causing students to struggle, not just list out the problems they're facing. That way, we can actually simulate interventions and see what might help.

For example, if we find that lack of transportation is one of the biggest issues, and that it's causing poor attendance, which leads to lower grades, maybe we can find a way to provide transportation. Otherwise, if I didn't know about the transportation problem, I might just push students into tutoring to fix their grades, without realizing that they physically can't even get to school.

Identify the **key stakeholders** for this scenario. How many distinct types of users do you think there are?

▶ 📌 **Click here to expand and see the scenario from Megan again** 📌

Write user stories for stakeholder 1. Each story should be on a separate line. Number each story.

Format: As a [type of user], I want [to do something], so that [I achieve this goal]. (you do not need to include the square brackets)

Example:

1. As a type of user, I want to do something, so that I achieve this goal.
2. As a type of user, I want to do something, so that I achieve this goal.
3. As a type of user, I want to do something, so that I achieve this goal.
4. etc.

Write acceptance criteria for each user story you created for stakeholder 1.

- Use the format: Given... when... then...
- You may write more than one criterion per user story.
- Label each criterion with the number of the user story it belongs to (e.g., 1., 2.,)
- Start each new criterion on a new line, even if a single criterion wraps onto multiple lines
- Put one criterion per line

Example Response:

1. Given I am completing hospital consent forms, when **I** am asked about the use of my medical data for educational purposes, then **I** must have a clear and separate option to either consent or decline**. (belongs to user story 1)**

1. Given I have opted in, when my data is accessed by the training system, then **I** must receive a notification in my patient portal that summarizes when and how it was used**. (belongs to user story 1)**

2. Given... when... then... **(belongs to user story 2)**

3. Given... when... then... **(belongs to user story 3)**

Label **each acceptance criteria for stakeholder 1** using the **MoSCoW method by adding one of the following tags at the beginning of each line.**

- Use **[M]** for Must
- Use **[S]** for Should
- Use **[C]** for Could
- Use **[W]** for Won't

You can **copy and paste your responses from the previous question** to get started.

Example:

1. [M] Given I am completing hospital consent forms, when **I** am asked about the use of my medical data for educational purposes, then I must have a clear and separate option to either consent or decline**. (belongs to user story 1)**

1. [S] Given **I** have opted in, when my data is accessed by the training system, then I must receive a notification in my patient portal that summarizes when and how it was

used. **(belongs to user story 1)**

2. [S] Given... when... then... **(belongs to user story 2)**

3. [C] Given... when... then... **(belongs to user story 3)**

▶ 📌 **Click here to expand and see the scenario from Megan again** 📌

Write user stories for stakeholder 2. Each story should be on a **separate line. Number each story.**

Format: As a [type of user], **I** want [to do something], so that [**I** achieve this goal]**.** (you do not need to include the square brackets)

Example:

1. As a type of user, I want to do something, so that **I** achieve this goal.
2. As a type of user, I want to do something, so that **I** achieve this goal.
3. As a type of user, I want to do something, so that **I** achieve this goal.
4. etc.

Write acceptance criteria for each user story you created for stakeholder 2.

- Use the format: Given... when... then...
- You may write more than one criterion per user story.
- Label each criterion with the number of the user story it belongs to (e.g., 1., 2.,)
- Start each new criterion on a new line, even if a single criterion wraps onto multiple lines
- Put one criterion per line

Example Response:

1. Given I am completing hospital consent forms, when I am asked about the use of my medical data for educational purposes, then I must have a clear and separate option to either consent or decline**. (belongs to user story 1)**

1. Given I have opted in, when my data is accessed by the training system, then I must receive a notification in my patient portal that summarizes when and how it was used. **(belongs to user story 1)**

2. Given... when... then... **(belongs to user story 2)**

3. Given... when... then... **(belongs to user story 3)**

Label **each acceptance criteria for stakeholder 2** using the **MoSCoW method by adding one of the following tags at the beginning of each line.**

- Use **[M]** for Must
- Use **[S]** for Should
- Use **[C]** for Could
- Use **[W]** for Won't

You can **copy and paste your responses from the previous question** to get started.

Example:
1. [M] Given I am completing hospital consent forms, when I am asked about the use of my medical data for educational purposes, then I must have a clear and separate option to either consent or decline**. (belongs to user story 1)**
1. [S] Given I have opted in, when my data is accessed by the training system, then I must receive a notification in my patient portal that summarizes when and how it was used**. (belongs to user story 1)**
2. [S] Given... when... then... **(belongs to user story 2)**
3. [C] Given... when... then... **(belongs to user story 3)**

▶ 📌 **Click here to expand and see the scenario from Megan again** 📌

Write user stories for stakeholder 3. Each story should be on a **separate line. Number each story.**

Format: As a [type of user], I want [to do something], so that [I achieve this goal]. (you do not need to include the square brackets)

Example:

1. As a type of user, I want to do something, so that I achieve this goal.
2. As a type of user, I want to do something, so that I achieve this goal.
3. As a type of user, I want to do something, so that I achieve this goal.
4. etc.

Write acceptance criteria for each user story you created for stakeholder 3.

- Use the format: Given... when... then...
- You may write more than one criterion per user story.
- Label each criterion with the number of the user story it belongs to (e.g., 1., 2.,)
- Start each new criterion on a new line, even if a single criterion wraps onto multiple lines
- Put one criterion per line

Example Response:

1. Given I am completing hospital consent forms, when I am asked about the use of my medical data for educational purposes, then I must have a clear and separate option to either consent or decline**. (belongs to user story 1)**

1. Given I have opted in, when my data is accessed by the training system, then I must receive a notification in my patient portal that summarizes when and how it was used**. (belongs to user story 1)**

2. Given... when... then... **(belongs to user story 2)**
3. Given... when... then... **(belongs to user story 3)**

Label **each acceptance criteria for stakeholder 3** using the **MoSCoW method by adding one of the following tags at the beginning of each line.**

- Use **[M]** for Must
- Use **[S]** for Should
- Use **[C]** for Could
- Use **[W]** for Won't

You can **copy and paste your responses from the previous question** to get started.

Example:
1. [M] Given I am completing hospital consent forms, when I am asked about the use of my medical data for educational purposes, then I must have a clear and separate option to either consent or decline**. (belongs to user story 1)**
1. [S] Given I have opted in, when my data is accessed by the training system, then I must receive a notification in my patient portal that summarizes when and how it was used**. (belongs to user story 1)**
2. [S] Given... when... then... **(belongs to user story 2)**
3. [C] Given... when... then... **(belongs to user story 3)**

▶ 📌 **Click here to expand and see the scenario from Megan again** 📌

Write user stories for stakeholder 4. Each story should be on a **separate line. Number each story.**

Format: As a [type of user], **I** want [to do something], so that [**I** achieve this goal]. (you do not need to include the square brackets)

Example:

1. As a type of user, I want to do something, so that **I** achieve this goal.
2. As a type of user, I want to do something, so that **I** achieve this goal.
3. As a type of user, I want to do something, so that **I** achieve this goal.
4. etc.

Write acceptance criteria for each user story you created for stakeholder 4.

- Use the format: Given... when... then...
- You may write more than one criterion per user story.
- Label each criterion with the number of the user story it belongs to (e.g., 1., 2.,)
- Start each new criterion on a new line, even if a single criterion wraps onto multiple lines

- Put one criterion per line

Example Response:
1. Given I am completing hospital consent forms, when I am asked about the use of my medical data for educational purposes, then I must have a clear and separate option to either consent or decline. **(belongs to user story 1)**
1. Given I have opted in, when my data is accessed by the training system, then I must receive a notification in my patient portal that summarizes when and how it was used**. (belongs to user story 1)**
2. Given... when... then... **(belongs to user story 2)**
3. Given... when... then... **(belongs to user story 3)**

Label **each acceptance criteria for stakeholder 4** using the **MoSCoW method by adding one of the following tags at the beginning of each line.**

- Use **[M]** for Must
- Use **[S]** for Should
- Use **[C]** for Could
- Use **[W]** for Won't

You can **copy and paste your responses from the previous question** to get started.

Example:
1. [M] Given I am completing hospital consent forms, when I am asked about the use of

my medical data for educational purposes, then I must have a clear and separate option to either consent or decline. **(belongs to user story 1)**

1. [S] Given I have opted in, when my data is accessed by the training system, then I must receive a notification in my patient portal that summarizes when and how it was used. **(belongs to user story 1)**

2. [S] Given... when... then... **(belongs to user story 2)**

3. [C] Given... when... then... **(belongs to user story 3)**

▶ 📌 **Click here to expand and see the scenario from Megan again** 📌

Write user stories for stakeholder 5. Each story should be on a **separate line. Number each story.**

Format: As a [type of user], I want [to do something], so that [I achieve this goal]. (you do not need to include the square brackets)

Example:

1. As a type of user, I want to do something, so that I achieve this goal.
2. As a type of user, I want to do something, so that I achieve this goal.
3. As a type of user, I want to do something, so that I achieve this goal.
4. etc.

Write acceptance criteria for each user story you created for stakeholder 5.

- Use the format: Given... when... then...
- You may write more than one criterion per user story.
- Label each criterion with the number of the user story it belongs to (e.g., 1., 2.,)
- Start each new criterion on a new line, even if a single criterion wraps onto multiple lines
- Put one criterion per line

Example Response:

1. Given I am completing hospital consent forms, when I am asked about the use of my medical data for educational purposes, then I must have a clear and separate option to either consent or decline**. (belongs to user story 1)**

1. Given I have opted in, when my data is accessed by the training system, then I must receive a notification in my patient portal that summarizes when and how it was used. **(belongs to user story 1)**

2. Given... when... then... **(belongs to user story 2)**

3. Given... when... then... **(belongs to user story 3)**

Label **each acceptance criteria for stakeholder 5** using the **MoSCoW method by adding one of the following tags at the beginning of each line.**

- Use **[M]** for Must
- Use **[S]** for Should
- Use **[C]** for Could
- Use **[W]** for Won't

You can **copy and paste your responses from the previous question** to get started.

Example:
1. [M] Given I am completing hospital consent forms, when I am asked about the use of my medical data for educational purposes, then I must have a clear and separate option to either consent or decline**. (belongs to user story 1)**
1. [S] Given I have opted in, when my data is accessed by the training system, then I must receive a notification in my patient portal that summarizes when and how it was used**. (belongs to user story 1)**
2. [S] Given... when... then... **(belongs to user story 2)**
3. [C] Given... when... then... **(belongs to user story 3)**

▶ 📌 **Click here to expand and see the scenario from Megan again** 📌

Based on the information you have from this scenario, imagine what the best AI system solution would be after an iterative development process. What kind of DATA TYPE would this system use?

Use the DATA TYPE section in the PDF to guide your decision.

- ○ Numeric
- ○ Tabular
- ○ Images
- ○ Audio
- ○ Time Series

Based on the information you have from this scenario, imagine what the best AI system solution would be after an iterative development process. What kind of LEARNING PARADIGM would this system use?

Use the LEARNING PARADIGM section in the PDF to guide your decision.

- ○ Supervised Learning
- ○ Unsupervised Learning
- ○ Reinforcement Learning
- ○ Semi-Supervised Learning
- ○ Imitation Learning
- ○ Human-in-the-loop Learning

Based on the information you have from this scenario, imagine what the best AI system solution would be after an iterative development process. What kind of MODEL TYPE/TECHNIQUE would this system use?

Use the MODEL TYPE/TECHNIQUES section in the PDF to guide your decision.

- ○ Graph Neural Networks
- ○ Transformers
- ○ Bandits
- ○ Clustering
- ○ Rule-Based

- ◯ Diffusion Models

Based on the information you have from this scenario, imagine what the best AI system solution would be after an iterative development process. What kind of SYSTEM BEHAVIOR/SETTING would this system have?

Use the SYSTEM BEHAVIOR/SETTING section in the PDF to guide your decision.

- ◯ Complementary Systems
- ◯ Cooperative Systems
- ◯ Fully Interactive Systems
- ◯ Decision Support Systems
- ◯ Autonomous Systems

Based on the information you have from this scenario, imagine what the best AI system solution would be after an iterative development process. What kind of TRAINING REGIME would this system use?

Use the TRAINING REGIME section in the PDF to guide your decision.

- ◯ Transfer Learning
- ◯ Few-Shot/Meta Learning
- ◯ Pre-Trained
- ◯ Trained from Scratch

Demographic Information

You've finished the stakeholder scenarios. Great work! Before we wrap up, we have a few quick demographic questions for you.

How old are you?

- ○ Less than 18 years old
- ○ 18 - 24 years old
- ○ 25 - 39 years old
- ○ 40 + years old

To which gender identity do you most identify?

- ○ Woman
- ○ Man
- ○ Non-binary
- ○ Genderqueer / Gender non-comforming
- ○ _____ Prefer to self-describe:
- ○ Prefer not to say

Which of the following best describes your race? Select all that apply

- ☐ American Indian or Alaska Native
- ☐ Asian
- ☐ Black or African American
- ☐ Hispanic or Latino
- ☐ Middle Eastern or North African
- ☐ Native Hawaiian or Pacific Islander
- ☐ White
- ☐ Prefer not to say

Which of the following best describes your **current** educational status?

- ☐ Undergraduate in Computer Science

- [ ] Undergraduate outside of Computer Science (please specify major)
- [ ] Master's student in Computer Science
- [ ] Master's student outside of Computer Science (please specify major)
- [ ] PhD student in Computer Science
- [ ] PhD student outside of Computer Science (please specify major)
- [ ] Not currently a student (graduated, working, etc.)

What college, university, or higher education institution are you currently studying at?

What degrees have you have **completed**? Multiple answers are permitted.

- [ ] Bachelor's degree in Computer Science
- [ ] Bachelor's degree in a field other than Computer Science (please specify major)
- [ ] Master's degree in Computer Science
- [ ] Master's degree in a field other than Computer Science (please specify major)
- [ ] PhD in Computer Science
- [ ] PhD in a field other than Computer Science (please specify major)
- [ ] None, I'm still pursuing my degree

Which of the following best describes your current employment? Multiple answers are permitted.

- [ ] Full/Part time job in a Computer Science related field in industry (Software engineer, Data scientist, etc.)

- [ ] Full/Part time job in a Computer Science related field in academia (Research assistant, Post-doc, etc.)
- [ ] Full/Part time job in another field in industry (please specify)
- [ ] Full/Part time job in another field in academia (please specify)
- [ ] I am self employed
- [ ] I am not currently working

How would you describe your background in Artificial Intelligence (AI)? Select all that apply.

- [ ] I have taken one or more AI-related courses (e.g., machine learning, computer vision, NLP)
- [ ] I have read academic papers or technical materials about AI
- [ ] I have implemented AI models or algorithms
- [ ] I have worked on applied AI projects
- [ ] I have formal training (degree, certification, etc.) in AI or a related field
- [ ] I have no background in AI
- [ ] Other

How would you rate your **theoretical understanding** of core AI concepts (e.g., machine learning, reinforcement learning, agents, optimization)?

- ○ No knowledge - I have not studied or encountered these concepts.
- ○ Basic familiarity - I've heard of some of these concepts or read brief overviews, but I couldn't explain them in detail.
- ○ Moderate understanding - I've learned about these topics in a course or from readings and could describe their purpose and how they work in general terms.
- ○ Strong understanding - I can explain most of these concepts in depth and understand when and why they are used.
- ○ Expert-level knowledge - I have a deep grasp of these concepts and could teach them or critique their use in technical or research contexts.

How would you rate your **practical experience** with implementing or applying core AI techniques (e.g., machine learning, reinforcement learning, agents, optimization)?

- ○ No experience - I have never implemented or applied AI methods.
- ○ Minimal experience - I've followed a tutorial or made small modifications to existing AI code, but not implemented anything from scratch.
- ○ Some experience - I've implemented AI techniques in coursework or personal projects but not in complex or real-world scenarios.
- ○ Substantial experience - I've applied AI methods in complex projects, research, or work settings, and I understand how to tune and troubleshoot them.
- ○ Expert-level experience - I have led AI development efforts, published implementation work, or significantly innovated on standard techniques.

Please select "strongly agree" for this question so we know you're paying attention.

- ○ Strongly Disagree
- ○ Disagree
- ○ Neutral
- ○ Agree
- ○ Strongly Agree

Have you ever conducted research involving AI? This includes any kind of project where you explored new ideas, built something to test a question, applied AI in a new setting, or worked on an AI-related topic to learn something new.

- ☐ No, I haven't worked on anything like that
- ☐ Yes, as part of a class project
- ☐ Yes, in an academic research lab
- ☐ Yes, in an industrial research lab
- ☐ Yes, as part of an industry or company project (not research-specific)
- ☐ Yes, at a hackathon or competition

- [ ] Yes, in a personal or independent project
- [ ] Other

Have you ever conducted or contributed to AI research or an AI project that aimed to address a social good problem (e.g., public health, climate, education, social justice, etc.)?

- [ ] No, I haven't worked on anything like that
- [ ] Yes, as part of a class project
- [ ] Yes, in an academic research lab
- [ ] Yes, in an industrial research lab
- [ ] Yes, as part of an industry or company project (not research-specific)
- [ ] Yes, at a hackathon or competition
- [ ] Yes, in a personal or independent project
- [ ] Other

For how many of the stakeholder scenarios did you use google or generative AI (e.g. ChatGPT or similar) to complete the question?

- ○ None
- ○ 1
- ○ 2
- ○ 3
- ○ 4
- ○ 5

What did you use google or generative AI (e.g. ChatGPT or similar) for?

How did you engage with the "AI System Components" PDF during the task?

- ○ I didn't look at the PDF at all
- ○ I glanced at it but didn't find it useful
- ○ I skimmed it and found a few helpful reminders of things I already knew
- ○ I referred to it multiple times and it helped clarify some concepts I wasn't sure about
- ○ I read it carefully because most of the concepts were new to me and it helped me complete the task
- ○ [ ] Other (please describe)

For how many of the four questions did you refer to the "AI System Components" PDF?

- ○ None
- ○ 1
- ○ 2
- ○ 3
- ○ 4

If you'd like to receive payment for your participation, please enter your email address below. The research team will review your responses and reach out with payment details if a good faith effort was shown throughout the study.

Your email will only be used to contact you about compensation.

Navigate to the next page to submit the survey. Your responses may not be counted if you do not submit.